

116 Advanced Statistics for Neuroscience

Exercise 03

Fitting curves

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Fitting a line and a parabola

Please write a Python programme that uses the supplied function "MakeObservations" to generate a set of observations x_i, y_i and fit these observations with both a line and a parabola, using either *curve_fit* from scipy or the formulas supplied in the lecture.

$$y(x_i) = a + bx_i, \qquad y(x_i) = a + bx_i^2$$

by using the above formulas.

In addition, compute the standard deviation of your best-fitting parameter estimates $\hat{a}$ and $\hat{b}$, as well as the standard deviation of observations y_i from model predictions $y(x_i)$, again using the above formulas.

Visualization and discussion

Plot your observations in an $x - y$ graph together with the two fitted curves (line and parabola).

Which curve (line or parabola) appears to provide a better fit, judging subjectively by eye? You may have to increase the number of observations to see a clear difference.

Which curve provides a better fit, judging objectively from the standard deviation of observations y_i ?

Which method appears to be more sensitive? Judging subjectively by eye, or objectively by standard deviation?

Why is it, do you think, that objective and subjective measures are so different?

Total, explained, and residual variation

Compute the total variation (sum of squared errors) in your observations!

Compute for each model (linear and parabolic) the explained variation and the residual variation!

Compute for each model the ratio of explained and residual variation!

Implementation hints

Importing

```
from scipy.optimize import curve_fit
```

and adding the following lines of code

```
# Define the first model function
```

```
def linear_model(x, a, b):
```

```
    return x * b + a
```

```
# Define the second model function
```

```
def parabolic_model(x, a, b):
```

```
    return x**2 * b + a
```

```
# Fit the two models
```

```
popt1, pcov1 = curve_fit(linear_model, x_i, y_i )
```

```
popt2, pcov2 = curve_fit(parabolic_model, x_i, y_i )
```